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Task-Technology Fit in Virtual Reality: Explaining Behavioral Intention to Use and Perceived Performance

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Abstract

While Virtual Reality (VR) is becoming more popular for professional applications, empirical evidence on its task-specific suitability remains scarce. This study employs the Task-Technology Fit (TTF) model to evaluate the perceived fit of VR for workplace tasks using McGrath's Group Task Circumplex. Results from an online survey were analyzed with Partial Least Squares Structural Equation Modeling (PLS-SEM). They indicate that TTF relates to performance perceptions and behavioral intention to use, explaining a substantial share of variance in both outcomes. The study links task taxonomies to TTF in immersive settings and shows that task and technology characteristics both contribute to perceived fit, with social presence the most salient technology factor. Practically, the findings support task-aware selection and design of VR solutions that align with task profiles. Limitations include simulated (non-headset) stimuli and a personalization-comparability trade-off introduced by task filtering.

Keywords: Virtual Reality, VR, Task-Technology Fit, TTF, McGrath Group Task Circumplex

1. Introduction

Immersive technologies such as Virtual Reality (VR) create digital environments that simulate real-world perception via computer-generated sensory input (Trindade et al., 2023; Wohlgenannt et al., 2020). These technologies are increasingly recognized as promising tools for supporting future digital work in organizations (e.g., Paetow et al., 2025). Despite VR technology having reached technical maturity and affordability for private use (Richter & Richter, 2023), its widespread adoption in professional settings remains limited (Lo et al., 2025; Wohlgenannt et al., 2020). Most empirical studies to date still focus on educational or experimental scenarios rather than routine workplace tasks (e.g., Paetow et al., 2025). However, practical applications, such as virtual collaboration and immersive onboarding (Trindade et al., 2023), demonstrate the relevance of VR in professional settings.

One challenge in assessing this potential is that new technologies are often assessed before actual use (e.g., Jeyaraj, 2022). Designs based on evaluative anticipation, in which participants assess technologies based on cognitive imagination rather than direct use, are an established method in early-stage research (Baxi et al., 2023; Jeyaraj, 2022).

The Task-Technology Fit (TTF) model explains how technology supports task performance (Goodhue & Thompson, 1995) and has been extended to pre-adoption contexts, where perceived fit relates to future usage (Al-Rahmi et al., 2020; Baxi et al., 2023).

Although McGrath's Group Task Circumplex is widely used to classify task types in group research (McGrath, 1984), its integration into quantitative structural models remains underdeveloped. We bridge this gap by embedding McGrath's taxonomy in a framework that enables empirical tests of task–technology relationships. Several task types (see Figure 1) conceptually align with VR's experiential affordances: Contest benefits from clear goals and feedback, Planning from visualization of abstract structures and future states, and Conflict from social presence, avatar interaction, and embodied cues (Dhar et al., 2023; Paetow et al., 2025; Yassien et al., 2020; Zhang et al., 2017). Thus, McGrath's taxonomy serves both as a measurement framework and a conceptual lens for anticipating task–technology fit in immersive environments.

Accordingly, we pose the following questions: First, to what extent do task characteristics defined by McGrath's taxonomy and technology characteristics relate to perceived TTF of VR in a professional context? Second, how does TTF relate to behavioral intention (BI) and perceived performance (PERF)?

To investigate this, we conducted an online survey with employed participants in Germany (n = 159), presenting VR as a prospective work tool via descriptive text and images. TTF was measured through task-

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relevance filtering, and the model was tested using Partial Least Squares Structural Equation Modeling (PLS-SEM), to allow higher-order constructs and direct evaluation of perceived fit.

Our findings confirm TTF as a key factor for both performance perceptions and usage intention in pre-adoption contexts. Because VR was introduced via descriptive stimuli rather than live experiences, validity and generalizability are limited. Future research should replicate with in-headset VR, consider cultural and organizational differences, and compare fit across immersive technologies, including Augmented and Mixed Reality, which may entail distinct user demands and design implications.

This paper is structured as follows: Section 2 outlines the theoretical background, including TTF, McGrath's taxonomy, and VR. Section 3 details the methodology and the PLS-SEM approach. Section 4 presents the results, followed by a discussion in Section 5. Finally, Section 6 concludes with implications, limitations, and directions for future research.

2. Theoretical Background

2.1. Task-Technology Fit (TTF)

The Task-Technology Fit (TTF) model, developed by Goodhue & Thompson (1995), posits that performance gains result from a high degree of alignment between a technology's characteristics and specific task requirements. Initially applied to enterprise systems, TTF has since evolved to encompass a broader range of user-centered technologies, including mobile services and digital platforms (Cane & McCarthy, 2009; Furneaux, 2012; Jeyaraj, 2022).

More recently, TTF has been used to examine immersive technologies such as VR, AR (e.g., Ho et al., 2023), and Virtual Worlds like Metaverses (e.g., Sharma et al., 2025). In these contexts, researchers emphasize that the perceived fit between immersive system capabilities (e.g., interactivity, presence, realism) and the cognitive or operational demands of the task is critical for user engagement and effectiveness.

For example, Zhang et al. (2017) demonstrate that in VR-supported learning environments, perceived TTF improves motivation and satisfaction when sensory immersion aligns with instructional needs. Khashan et al. (2023) confirm these effects in higher education, where TTF relates to both satisfaction and intention to continue using immersive platforms, particularly under high task clarity and compatibility demands.

Furthermore, Spies et al. (2020) note that TTF is often applied in early-stage or theoretical evaluations, especially when empirical usage data is lacking. This highlights its suitability as a framework for assessing expected fit and user alignment in emerging technologies such as VR.

2.2. McGrath's Group Task Circumplex

A structured understanding of task characteristics is essential for analyzing technology-supported systems within the TTF model (Goodhue & Thompson, 1995) and addresses repeated calls for more systematic approaches to modeling task constructs in IS research (e.g., Chavarnakul et al., 2024).

One framework for describing task characteristics is McGrath's (1984) Group Task Circumplex, which classifies tasks along two key dimensions: (1) cognitive vs. behavioral orientation and (2) level of interdependence and coordination (see Figure 1).

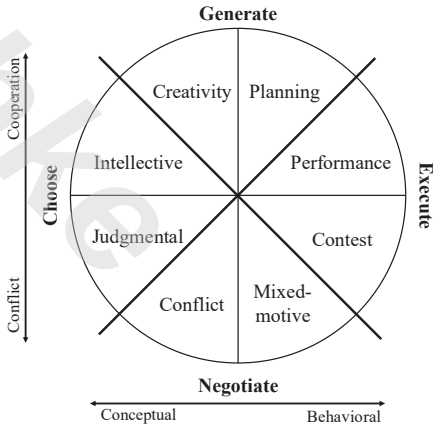


Figure 1. Group Task Circumplex (McGrath, 1984).

This framework yields four task categories, each with two subtypes (McGrath, 1984):

Generate tasks include planning tasks, which aim to develop coherent action plans, and creativity tasks, which focus on divergent thinking and innovation. Both task types typically involve low levels of coordination. Choose tasks comprise intellective tasks, which require objectively correct solutions, and judgmental tasks, which involve subjective decisions among multiple alternatives. Judgmental tasks generally demand higher levels of coordination. Negotiate tasks consist of cognitive conflict tasks (e.g., differing interpretations of information) and mixed-motive tasks, which involve underlying conflicts of interest. These tasks are typically highly interdependent and characterized by conflict. Execute tasks include contest tasks, which emphasize competition (e.g., performance comparisons), and performance tasks, which require precise, coordinated physical execution under defined conditions.

Although originally developed to analyze group activities, the model can be applied beyond collaborative settings. Zigurs & Buckland (1998) argue

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that structural complexity, not necessarily group dynamics, is the key determinant of technology fit. Straus (1999) further validates the framework's vertical dimension, demonstrating that interdependence is correlated with coordination effort. Moreover, Wilson & Connolly (2001) demonstrate that users can meaningfully evaluate different task types and technology suitability depending on context.

2.3. Virtual Reality (VR)

Virtual Reality (VR) refers to fully computer-generated environments that replace real-world sensory input with synthetic stimuli such as 3D visuals, spatial audio, and haptic feedback (Trindade et al., 2023; Wohlgenannt et al., 2020).

VR is increasingly utilized in domains such as healthcare, industrial training, and education, particularly for tasks requiring spatial cognition, procedural knowledge, or collaboration under physical or economic constraints (Dhar et al., 2023; Paetow et al., 2025; Trindade et al., 2023). In professional contexts, typical use cases include onboarding, design reviews, and safety training (Trindade et al., 2023).

Studies have shown that training in immersive environments can enhance learning outcomes, user confidence, and self-efficacy (Dhar et al., 2023), while also contributing to improved mental well-being and physical safety at work (Paetow et al., 2025).

To explain how users evaluate immersive technologies such as VR, three core characteristics are frequently considered: immersion, interactivity, and (social) presence (Walsh & Pawlowski, 2002; Wohlgenannt et al., 2020; Yassien et al., 2020). Immersion refers to the psychological sensation of being fully immersed in a mediated environment driven by multisensory stimuli (Witmer & Singer, 1998). Interactivity describes the user's perceived control and influence over system behavior, often realized through real-time responsiveness and bidirectional feedback (Liu, 2003). Social presence refers to the extent to which users perceive others as being physically or socially "real" within the virtual environment, facilitated through avatars, gestures, and synchrony (Poeschl & Doering, 2015). This dimension is particularly relevant for work-related tasks that involve communication, emotional understanding, or interpersonal dynamics. Research shows that social presence can influence collaboration, emotion recognition, and perceived rapport, even in task-focused virtual settings (Sun & Won, 2021).

Although VR still faces challenges such as latency and cybersickness, recent technological advancements have improved its robustness and enterprise applicability, particularly in domains requiring spatial cognition, procedural training, or teamwork (Dhar et al., 2023; Paetow et al., 2025).

2.4. Related Work on the Effects of TTF

The TTF model has been extensively examined over the past few decades as a strong factor for performance across a wide array of technologies and domains. Meta-analyses confirm that a high perceived fit between task requirements and technological features enhances decision quality, task completion, and productivity (Cane & McCarthy, 2009; Chavarnakul et al., 2024; Jeyaraj, 2022). These effects are particularly pronounced when TTF is assessed directly via self-report rather than calculated as an interaction construct (e.g., Cane & McCarthy, 2009).

Beyond performance, TTF has been shown to relate to behavioral intention to use technology, particularly in early-stage or hypothetical scenarios (e.g., Jeyaraj, 2022). Studies in immersive learning environments (Zhang et al., 2017), retail (Khashan et al., 2023), MOOCs (Ulfa et al., 2024), and blockchain applications (Patil et al., 2022) report that anticipated fit (users' subjective expectation of usefulness) shapes adoption willingness, even in the absence of usage experience. Similar effects are reported in e-learning contexts, where perceived alignment between system features and learning needs enhances both intention and satisfaction (Al-Rahmi et al., 2020). Supporting these findings, Baxi et al. (2023) demonstrate that TTF relates to digital wallet adoption, even when accounting for core Unified Theory of Acceptance and Use of Technology (UTAUT) variables such as performance expectancy. These results underline the motivational value of perceived fit in technology acceptance, especially in uncertain or pre-use scenarios.

In emerging technologies like VR, TTF captures the alignment between task demands and experiential features such as immersion, interactivity, and presence. This alignment has been linked to stronger engagement and motivational outcomes (Spies et al., 2020; Zhang et al., 2017).

2.5. Hypotheses & Research Model

Extensive empirical and meta-analytic evidence demonstrates a positive relationship between perceived TTF and individual outcomes such as task execution, decision quality, and satisfaction across diverse domains (Cane & McCarthy, 2009; Chavarnakul et al., 2024; Jeyaraj, 2022). In immersive contexts, this assumption remains relevant.

Studies highlight that when technology is perceived to align with complex task requirements, users report greater confidence in achieving their goals and greater

perceived performance outcomes – both in immersive (Chavarnakul et al., 2024; Zhang et al., 2017) and educational online settings (Ulfa et al., 2024). Therefore, we propose:

H1: TTF is positively associated with perceived individual performance with VR.

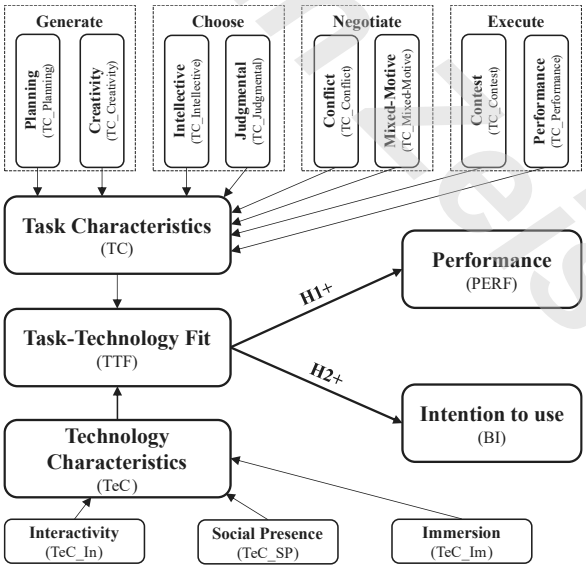


Figure 2. Research model.

TTF has also been repeatedly shown to correlate with users' intention to adopt new technologies even before actual usage occurs. In pre-adoption or hypothetical scenarios, perceived fit functions as a cognitive proxy for anticipated usefulness. Chavarnakul et al. (2024) and Al-Rahmi et al. (2020) report strong effects on BI in immersive and educational contexts. Therefore, we posit:

H2: TTF is positively associated with the intention to use VR.

Figure 2 summarizes the proposed research model. Task Characteristics (TC) follow McGrath's taxonomy (see Section 2.2), while Technology Characteristics (TeC) refer to the characteristics of VR (see Section 2.3).

3. Methods

3.1. Survey Design

To test the proposed hypotheses, we conducted a structured online survey (see Appendix, Survey). The survey included 60 closed-ended items across five constructs: Task Characteristics (TC), Technology Characteristics (TeC), Task-Technology Fit (TTF), Behavioral Intention (BI), and Perceived Performance (PERF).

TC was measured with 24 items (3 per task type) based on McGrath's (1984) Group Task Circumplex, using a six-point frequency scale (1 = "never", 6 = "daily"). TeC was assessed using 9 items (3 each for immersion: TeC_Im, interactivity: TeC_In, and social presence: TeC_SP) adapted from established scales (Poeschl & Doering, 2015; Witmer & Singer, 1998). BI (3 items) was adapted from UTAUT (Venkatesh et al., 2003), and PERF (3 items) from the original TTF model (Goodhue & Thompson, 1995).

To ensure valid assessments, TTF was measured using a relevance-based filtering procedure: only those task types reported as performed at least monthly (response ≥ 4 on the 6-point frequency scale) were followed by corresponding fit items. For example, if a participant indicated that they performed "planning" tasks monthly or more often, they would then rate how well VR supports this task. Tasks performed less frequently were excluded from the fit evaluation. While this method enhances validity and reduces respondent burden by focusing participants only on tasks they actually perform, it also reduces variance because each participant evaluates a different, personalized subset of task types (Gribbons & Hocevar, 1998; Schuessler et al., 2025). This individualized approach improves contextual relevance but limits comparability across participants, as not all subjects respond to the same TTF items.

All constructs except TC were measured on a five-point Likert scale (1 = "strongly disagree" to 5 = "strongly agree"). VR was introduced as a prospective work tool through descriptions that used text and images (see Appendix, Survey).

At the first-order level, all latent constructs were specified reflectively. A pretest (n = 50) evaluated item clarity and filter logic. According to Goodman et al. (2013), two attention checks and one control question were included to ensure data quality.

3.2. Data Collection & Participants

Data were collected in Germany in May and June 2025 via Clickworker, a crowdsourcing platform offering access to a diverse respondent pool. Since the target group comprised employed individuals in Germany, the survey was conducted in German.

All procedures adhered to the ethical standards outlined in the 1964 Helsinki Declaration and its subsequent amendments (World Medical Association, 2013). Participants were recruited anonymously, provided informed consent, and received monetary compensation in accordance with the German minimum wage. Since no personal or sensitive data were collected and no intervention occurred, no formal ethics approval was required.

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Out of 489 initial respondents, 383 completed the questionnaire, resulting in a completion rate of 78.3%. After applying exclusion criteria based on attention and control checks, 159 valid responses remained, representing 41.5% of the total. This sample comprised 96 males (60.4%) and 63 females (39.6%) aged 18 to 63 years (M = 39.6, SD = 11.6, Median = 39.0). The largest age group was 36 to 45 years (36.3%).

3.3. Model Specification & Data Preparation

To test the proposed research model and our two hypotheses (see Figure 2), we employed PLS-SEM using the software SmartPLS 4.1.1.2 (Ringle et al., 2024). PLS-SEM was selected due to its robustness in estimating complex models involving multiple latent variables and its ability to handle non-normal data distributions (Chin, 1998; Hair et al., 2022; Hair et al., 2019). The model was initially specified as reflective-formative: TC and TeC were modeled as Type II higher-order constructs composed of reflective first-order dimensions (e.g., TC_Conflict, TC_Planning, etc.) (Sarstedt et al., 2019). All first-order constructs were assessed for indicator reliability, internal consistency, and convergent validity following established criteria (Hair et al., 2022). The item TC_Performance_3 was removed due to low and non-relevant loading, consistent with best practices (Hair et al., 2022).

Subsequently, the model was reconfigured using latent variable scores (LVS). This transformation enabled both TC and TeC to be modeled as formatively aggregated, unidimensional constructs, providing a pragmatic estimation strategy in settings involving construct complexity (Sarstedt et al., 2019). During this step, the LVS indicators Planning, Judgmental, Performance, and Intellective were excluded due to insufficient loading or significance, resulting in a final TC construct comprising the four reliable dimensions: Conflict, Contest, Creativity, and MixedMotive. To mitigate parameter bias associated with reflectively specified models, we applied the Consistent PLS (PLSc) algorithm (Henseler et al., 2024). The relevance of path coefficients was tested using consistent PLS-SEM bootstrapping with 10,000 subsamples (e.g., Aguirre-Urreta & Rönkkö, 2018) of bias-corrected and accelerated confidence intervals (Hair et al., 2019).

TTF was modeled as a participant-specific single-indicator variable, calculated in IBM SPSS Statistics as the mean of all TTF items linked to task types that were objectively deemed relevant based on a minimum execution frequency (≥ monthly; see Section 3.1).

Finally, a sensitivity analysis using G*Power 3.1.9.7 (see Appendix) indicated that, for a sample size of n = 159, $\alpha = .05$, and 5 predictors, the minimum detectable effect size at 80% power was $f^2 = 0.0836$ (Cohen, 2013). This threshold was used to contextualize observed effect sizes, following PLS-SEM-specific guidance (Kock & Hadaya, 2018).

3.4. Data Analysis

Following Hair et al. (2019), the measurement model was evaluated in four steps. First, indicator reliability was confirmed, with all outer loadings for reflective constructs exceeding the 0.708 threshold (see Appendix, Table 1). Internal consistency was assessed via rho_A, with all values above 0.708 (see Appendix, Table 2). Convergent validity was supported by Average Variance Extracted (AVE) values > 0.50 (see Appendix, Table 2), and discriminant validity was established through the Heterotrait-Monotrait Ratio (HTMT), with all values below 0.90 (see Appendix, Table 3). For formative constructs (TC, TeC), variance inflation factor (VIF) values remained below 3.0, indicating no critical multicollinearity. However, Bootstrapping revealed that only TC_Contest and TeC_SP had relevant weights ($p < .001$). All other indicators, though low relevance in weight, showed relevant loadings ($p < .01$), justifying their retention based on absolute contribution (Hair et al., 2019) (see Appendix, Table 4).

Following the assessment of the measurement model, the structural model was evaluated (Hair et al., 2019). All inner VIF values for predictor constructs were below 3.3, indicating no evidence of multicollinearity within the structural model (see Appendix, Table 5). The model explained 52.0% of the variance in BI and 55.9% in PERF (see Appendix, Table 6), both of which represent moderate levels of explanatory power (Hair et al., 2019). The R^2 for TTF was 17.7%, which is considered weak, yet acceptable given the reduced variance in the TTF construct resulting from conditional weighting in its scoring procedure. Path coefficients between TTF and the two endogenous variables were substantial, as reflected in the corresponding effect sizes (see Appendix, Table 7): both relationships yielded large effects in line with Cohen's classification (Cohen, 2013).

Predictive performance was assessed using PLSpredict (10-fold cross-validation, 10 repetitions) and Cross-Validated Predictive Ability Testing (CVPAT) (Shmueli et al., 2019). All Q^2_predict values were above zero (see Appendix, Table 9), indicating predictive relevance compared to a naïve benchmark (see Appendix, Table 8). As the items of BI and PERF were not highly symmetrically distributed, we report on PLS-MAE values instead of PLS-RMSE. At the indicator level, the PLS model showed higher MAE than the linear model (LM) benchmark for most items, particularly for TTF and PERF (see Appendix, Table 9).

This results in no predictive accuracy of the model in terms of early incidents. On the construct level, MAE values for all latent variables were lower in the PLS model than in the indicator average (IA) benchmark, and significantly so for PERF ($p = .012$; see Appendix, Table 9). No significant advantage was found over the LM benchmark (all $p > .05$; see Appendix, Table 10). At the construct level, the PLS model outperformed the IA benchmark, with significantly lower prediction error for PERF ($\Delta MAE = -0.173$, $p = .012$). CVPAT results further confirmed this, showing a significantly lower average prediction error for the overall model compared to the IA benchmark ($\Delta MAE = -0.136$, $p = .032$). No significant difference was found compared to the LM benchmark ($p = .918$), providing partial support for the model's out-of-sample predictive validity (Shmueli et al., 2019).

Model fit was evaluated via the Standardized Root Mean Square Residual (SRMR). Following Hair et al. (2019), values below 0.08 indicate acceptable fit. The SRMR was 0.079 for the estimated model and 0.050 for the saturated model, supporting overall model adequacy (see Appendix, Table 11).

4. Results

The model accounts for 52.0% of the variance in behavioral intention to use (BI) and 55.9% in perceived performance (PERF) (see Figure 3), representing moderate explanatory power in line with the recommendations by Hair et al. (2019). For Task-Technology Fit (TTF), 17.7% of the variance is explained, falling below the typical 0.25 threshold for weak explanatory power (Hair et al., 2019).

Both hypotheses are supported by the structural model (see Figure 3). TTF exerts a strong and relevant relation to both BI ($\beta = 0.836$, $p < .001$, 95% CI [0.714, 0.957], $f^2 = 0.523$) and PERF ($\beta = 0.810$, $p < .001$, 95% CI [0.701, 0.923], $f^2 = 0.531$), confirming large effect sizes. These findings affirm the central proposition of the TTF model (Goodhue & Thompson, 1995): greater perceived alignment between task and technology relates to stronger performance expectations (even under pre-adoption conditions).

Concerning the antecedents of perceived fit, both TC ($\beta = 0.312$, $p < .001$, 95% CI [0.161, 0.418], $f^2 = 0.116$) and TeC ($\beta = 0.242$, $p = .002$, 95% CI [0.099, 0.362], $f^2 = 0.070$) relates to TTF.

No relevant effects were observed for the moderating roles of age and gender. The relationships between TTF, BI, and PERF thus appear robust across demographic subgroups, with no evidence of moderation.

The Pearson correlation coefficients between the constructs are presented in the Appendix (Table 12).

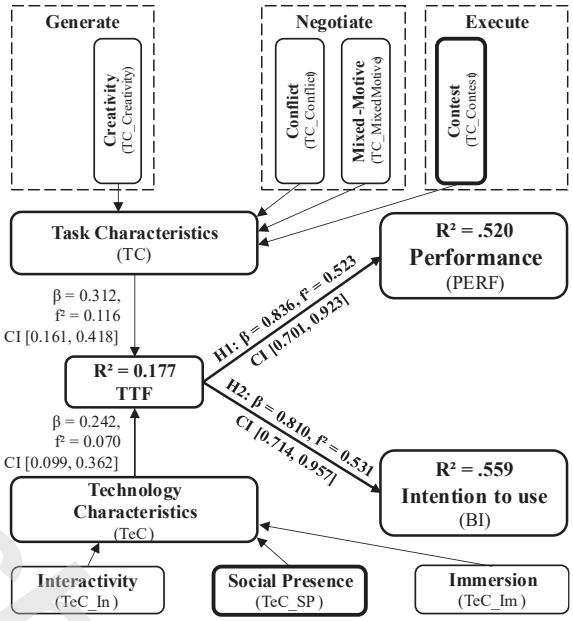


Figure 3. Research model results (one-tailed).

5. Discussion

This study investigated how task-technology fit (TTF) relates to perceived performance (PERF) and behavioral intention to use (BI) virtual reality (VR) for work tasks in a pre-adoption context.

TTF demonstrated a strong correlation with perceived performance, confirming its core proposition: aligning task demands with technological capabilities enhances users' confidence in their ability to perform tasks effectively (Goodhue & Thompson, 1995). Well-fitting technologies have long been linked to better task outcomes. In VR, this occurs through sensory fidelity and real-time responsiveness, which promote engagement and flow during task execution (Zhang et al., 2017).

Beyond performance, TTF also strongly relates to users' behavioral intention to use VR. This confirms prior findings from MOOCs, retail, and immersive learning, where perceived fit is key to adoption in early-stage or hypothetical contexts (Al-Rahmi et al., 2020; Khashan et al., 2023; Ulfa et al., 2024). Baxi et al. (2023) found that TTF relates to intention to use digital wallets even when controlling for UTAUT variables such as performance expectancy, suggesting that perceived fit exerts a unique motivational pull in uncertain or pre-use situations. These findings are also consistent with UTAUT-based logic, in which fit functions as a cognitive antecedent of performance expectancy (Venkatesh et al., 2003). Together, these results confirm TTF's dual role as a performance and motivation driver in pre-adoption settings.

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The findings suggest that even before implementation, perceived fit can guide early-stage design decisions, and early evaluations such as prototype testing and scenario walkthroughs help identify which features users find helpful or motivating. Studies confirm that imagined or hypothetical fit is reliably relating to intention and preference (Al-Rahmi et al., 2020; Khashan et al., 2023). Practical examples from onboarding and VR training show how structured tasks support this process (Wohlgenannt et al., 2020). Planning with fit in mind early and step by step can shape successful system development, and practitioners should incorporate scenario-based prototypes early to assess perceived fit with real tasks.

However, the future acceptance of immersive systems may depend on more than perceived fit alone. Potential barriers, such as cybersickness, privacy concerns, and a lack of organizational support, may reduce real-world usage, even with positive expectations (e.g., Paetow et al., 2025). As prior studies have shown, access to VR hardware alone is insufficient; actual use depends on integration into workflows, user confidence, and available time resources (Lo et al., 2025; Wohlgenannt et al., 2020). Although not central to TTF, these factors should be integrated into future extensions for realistic adoption modeling.

Regarding the antecedents of fit, both Technology Characteristics (TeC) and Task Characteristics (TC) highly relate to TTF. This underscores that users evaluate fit through both what they do and how technology supports it.

Among the technology characteristics (TeC), social presence (TeC_SP) had the strongest weight. This suggests that interactive and responsive features, such as avatars, gestures, or spatial audio, are essential for users' sense of fit.

Prior studies show that even simple avatars can convey emotional and social cues through gestures and proximity, which supports perceived alignment (Sun & Won, 2021). In persistent virtual environments – Metaverses – VR serves as a key interface for collaboration and shared task relevance (Paetow et al., 2025; Richter & Richter, 2023; Sharma et al., 2025). In these shared and persistent spaces, social presence is central to collaboration and task relevance. Features such as gaze, gesture, and synchrony foster co-presence, which in turn enhances perceived fit (Sun & Won, 2021; Yassien et al., 2020). Designers should prioritize features that strengthen emotional connection such as embodiment, spatial audio, and real-time responsiveness rather than focusing solely on technical usability. These features can also help address psychological needs, such as reducing isolation in virtual teamwork (Paetow et al., 2025).

The relevant effect of Task Characteristics (TC) suggests that users' perceived fit with VR systems depends not only on technical features but also on the structure and demands of the tasks themselves. Among the task-related aspects, the "Contest" dimension contributed the most. This means that goal-oriented, competitive activities may help users more easily imagine how VR could support their work. While this is based on relative weight and should be interpreted with caution, it suggests that clear goal structures may help users imagine VR use more concretely. This aligns with prior work showing that Contest-type tasks conceptually match VR's strengths in providing structured feedback, real-time interaction, and performance-focused engagement (Dhar et al., 2023; Paetow et al., 2025; Zhang et al., 2017). Likewise, Conflict-type tasks, although less dominant empirically, are theoretically relevant because VR can leverage avatar-based embodiment and social presence to support negotiation and discussion (Yassien et al., 2020). This demonstrates that McGrath's framework adds more than operational variety: it helps explain why particular task types conceptually align with immersive technologies.

At the same time, the prominence of Contest may also reflect cultural and contextual factors. German workplaces are often characterized as highly achievement-oriented and competitive, which could have amplified the salience of contest-type tasks in our sample (e.g., Brodbeck et al., 2002). In addition, recent research shows that immersive technologies themselves can reinforce competitive mechanisms, thereby further increasing the perceived relevance of performance-oriented activities (e.g., Zhan et al., 2025). Together, this underscores the need for future studies to examine whether Contest emerges as equally dominant in other cultural contexts or when VR scenarios emphasize collaboration rather than competition.

Overall, the model explained 52.0% of the variance in BI and 55.9% in PERF, representing substantial explanatory power in line with recommendations for PLS-SEM (Hair et al., 2019). In contrast, the explained variance of TTF was lower ($R^2 = 17.7\%$). This weaker value does not necessarily indicate a model flaw but may reflect the use of a relevance-based filtering procedure for TTF. By restricting the evaluation to tasks that are frequently performed individually, the approach enhances contextual validity while naturally reducing variance across participants (Gribbons & Hocevar, 1998; Schuessler et al., 2025). However, this personalization introduces a methodological trade-off: it reduces psychometric uniformity, since not all participants evaluate the same items. This limits cross-participant comparability and scale-based generalizability. Future studies may enhance this understanding by modeling task types separately.

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Furthermore, no relevant moderating effects of age or gender were found. However, this finding may also be influenced by the characteristics of the sample, German employees. It may not fully capture effects that could arise in other cultural or organizational contexts.

In summary, this study affirms TTF as a strong factor to relate to both BI and PERF in VR contexts, even under pre-adoption conditions. Both task and technology drivers are highly relevant and offer a theory-grounded basis for early-stage VR evaluation. By identifying design and implementation factors that enhance perceived fit, organizations can increase the likelihood that early VR initiatives translate into meaningful, task-relevant use. This highlights the practical value of using perceived fit as a design compass in early-phase innovation work, particularly when long-term behavioral data is not yet available. Nevertheless, without organizational readiness and user support, even promising VR concepts may fail to succeed unless they are integrated into organizational routines and receive user-centered support. Future research should expand the model by incorporating individual factors such as self-efficacy and by investigating whether the observed fit dynamics also apply to other immersive technologies such as augmented or mixed reality, which may involve different design requirements.

6. Conclusion

6.1. Theoretical Implications

This study contributes to the theoretical advancement of the TTF model in VR as an immersive environment in several ways.

First, the findings indicate that TTF relates not only perceived performance, as suggested by Goodhue & Thompson's original model, but also to behavioral intention to use VR. This dual role highlights TTF's value as both a motivational and performance-oriented construct in pre-adoption settings.

Second, the study applies McGrath's Group Task Circumplex through a formative second-order model. Integrating task demands, such as judgment, creativity, and conflict, into a unified framework provides a theory-driven lens for modeling task complexity in IS research.

Lastly, both task and technology characteristics mainly related to TTF. Social presence was the strongest technological factor, emphasizing the importance of relational cues, while the task dimension "Contest" stood out as particularly relevant. These results suggest a cognitive-emotional dimension of TTF, correlating with the nature of tasks and immersive experiences.

6.2. Practical Implications

This study provides several practical implications for the development and implementation of VR systems in workplace contexts.

First, the strong effects of TTF on performance and intention highlight that perceived fit should guide early-stage design decisions. Scenario-based prototypes and walkthroughs can be used to evaluate how well a VR concept aligns with real work tasks, even before technical development begins.

Second, features that enhance social presence such as embodiment, avatar gaze, spatial audio, and synchrony should be prioritized, especially in collaborative settings. These relational features can lead to higher users' perceived alignment and emotional engagement more than technical fidelity alone.

Third, the relevance of task characteristics, particularly the dimension "Contest" (TC_Contest), suggests that goal-driven and competitive activities may be especially suitable for VR adoption. Organizations should consider structuring early use cases around clear task goals and performance outcomes.

Finally, for long-term success, technical fit must be accompanied by organizational readiness. This includes reducing adoption barriers (e.g., cybersickness, privacy concerns), offering gradual onboarding, and integrating VR into real workflows. Simple adjustments like lowering motion intensity and allowing user control can support sustainable use.

6.3. Limitations & Future Research

This study has several limitations that also point to opportunities for future research.

First, participants were shown VR as description using text and images rather than experiencing it directly. While suitable for early-stage research, this approach cannot fully capture embodied interaction. Core experiential features such as spatial presence and sensory feedback were not represented, potentially limiting how vividly participants imagined fit. Future studies should replicate the model using live VR systems in lab or field settings to improve validity.

Second, the sample consisted of German-speaking Clickworker users, which may introduce cultural and self-selection bias. As attitudes toward immersive technologies vary by context, cross-cultural and cross-sector replications are needed to assess generalizability.

Third, negative effects of VR such as cybersickness, sensory overload, or fatigue were not examined. These issues may not only reduce acceptance but also impair perceived performance, even when task-technology fit is high. Future studies should explore

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how design factors (e.g., frame rate, navigation, interaction pacing) can mitigate such effects.

Fourth, this study focused on VR, but the model could apply to other immersive technologies like AR or MR. Comparative studies are needed and could clarify whether similar fit drivers such as social presence apply across environments.

Lastly, relevance-based filtering produced personalized TTF scores, improving task relevance but reducing comparability. To balance ecological validity and measurement consistency, future work should compare filtered and unfiltered scoring and, where appropriate, model task types separately or control task exposure.

7. Appendix

Available in Open Science Framework (OSF) at: https://osf.io/ep23c/?view_only=49136780e01a44deb91b0cf7bd71772a.

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